Academic Office

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

M.Tech Programmes

Academic Regulations (Highlights)

M.Tech in Computer Science and Engineering
M.Tech in Computer Science and Information Security (Applicable from 2022 batch)
M.Tech in Computer Aided Structural Engineering
M.Tech in Product Design and Management (Applicable from 2024 batch)
M.Tech in VLSI & Semiconductor Technologies (Applicable from 2026 batch)

M.Tech CSE - Academic Regulations (Highlights)

Credit Requirements:

- Minimum credits required for graduation is 66. Each semester, every student must register for at least 12 credits and at most 20 credits, which can include full or half courses.
- Every student must register for 16 *bouquet core elective* credits. CSE bouquet core courses belong to 3 categories of Theory, Systems and AI. A student must select 1 course from each of these 3 categories, and 1 additional course in any of these 3 categories, adding to a total of 4 bouquet courses. These courses are usually offered at least once every year.
- Every student must register for at least 20 *area elective* credits from the broad IT area (CSE/ECE). For every elective at the 400+ level, the area under which it belongs shall be made available before registration every semester. Area electives include bouquet core electives.
- Every student must register for 4 *breadth elective* credits, which include engineering, science & math electives offered in the institute. Breadth electives include area electives.
- Every student must register for 8 credits of *open electives*, which include any 400+ level elective offered in the institute. Open electives include breadth electives.

Bouquet Core Electives

- **Theory:** Modern Complexity Theory, Advanced Algorithms, Optimization Methods, Principles of Programming Languages, Principles of Information Security.
- **Systems:** Advanced Computer Networks, Data Systems, Distributed Systems, Compilers, Software Engineering
- **AI:** Statistical Methods in AI, Information Retrieval and Extraction, Data Analytics I, Computer Vision, Advanced NLP

Project Work:

- At most 4 credits can be converted to project / independent study credits in lieu of courses.

Academic Performance:

- A student should complete the requirements with a minimum CGPA of 6.00 to receive the M.Tech degree.

Residency Requirements:

- **Full-time students:** Students will have minimum of 4 semesters and maximum of 6 semesters to complete the graduate requirements, failing which they will be terminated from the programme.
- **Part-time students:** Students will have minimum of 4 semesters and maximum of 8 semesters to complete the graduate requirements, failing which they will be terminated from the programme.

Fees:

- The student has to pay full-time post-graduate fees for the first 4 semesters of study. The fees will be pro-rated to the number of credits registered for thereafter according to the institute's policies.

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M.Tech CSIS Academic Regulations (Highlights)
(Applicable from 2022 batch)

Credit Requirements:

- Minimum credits required for graduation is 66. Each semester, every student must register for at least 12 credits and at most 20 credits, which can include full or half courses.
- Every student must register for 16 area credits out of which 4 credits of project in 4th Semester.
- Every student must register for 20 credits of Bouquet electives. Of these 2 courses namely Advanced Computer Networks and Principles of Information Security are mandatory. Of the remaining 3, at least 1 must be from the AI Bouquet
- Every student must register for 12 credits of *open electives*, which include any 400+ level electives offered in the institute. Open electives include breadth electives.

Bouquet Electives

- **Theory:** Modern Complexity Theory, Advanced Algorithms, Optimization Methods, Principles of Programming Languages, Principles of Information Security.
- **Systems:** Advanced Computer Networks, Data Systems, Distributed Systems, Compilers, Software Engineering
- **AI:** Statistical Methods in AI, Information Retrieval and Extraction, Data Analytics I, Computer Vision, Advanced NLP

Project Work:

- Every student must register for 4 credits of project in 4th semester under C-STAR faculty.

Academic Performance:

- A student should complete the requirements with a minimum CGPA of 6.00 to receive the M.Tech degree.

Residency Requirements:

- **Full-time students:** Students will have minimum of 4 semesters and maximum of 6 semesters to complete the graduate requirements, failing which they will be terminated from the programme.

- **Part-time students:** Students will have minimum of 4 semesters and maximum of 8 semesters to complete the graduate requirements, failing which they will be terminated from the programme.

Fees:

- The student has to pay full-time post-graduate fees for the first 4 semesters of study. The fees will be pro-rated to the number of credits registered for thereafter according to the institute's policies.

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M. Tech in Computer Aided Structural Engineering
Academic Regulations (Highlights)

Credit Requirements:

- Student has to acquire a minimum of 66 credits in 4 semesters to become eligible to receive M. Tech in CASE degree. This can be achieved by acquiring credits as follows:
    - Minimum 24 credits from Structural Engineering area
    - Minimum 12 credits from Computer Science and IoT
- 18 credits from any SE or other electives of their choice (recommended minimum of 4 credits from each bucket)
    - 12 credits from project work/internship in 4th semester
- **Student can register a maximum of 20 credits either in 2nd or 3rd semester.**
- **Student can register up to a maximum of 2 Independent study courses in 2nd or 3rd semester.**

Project Work/Internship:

- Students are encouraged to do summer project/internship for 12 credits at IIIT Hyderabad or in construction industry/design consultancy.

Grading Scheme

- The grading for all courses will follow letter grading scheme with grade points stipulated by the institute.
- Project/internship completed in fourth semester (for 12 credits) will follow Satisfactory/Unsatisfactory grading scheme without grade points.

Academic Performance:

- A student should complete the requirements with a minimum CGPA of 6.00 to receive the M.Tech degree.

Residency Requirements:

- **Full-time students:** Students will have minimum of 4 semesters and maximum of 6 semesters to complete the graduate requirements, failing which they will be terminated from the programme.

- **Part-time students:** Students will have minimum of 4 semesters and maximum of 8 semesters to complete the graduate requirements, failing which they will be terminated from the programme.

Fees:

- The student has to pay full-time post-graduate fees for the first 4 semesters of study. The fees will be pro-rated to the number of credits registered for thereafter according to the institute's policies.

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M.Tech in Product Design and Management
Academic Regulations (Highlights)
(Applicable from 2024 batch)

Credit Requirements:

A total of 66 credits must be completed.

- Core courses - 22 credits
- Design electives – Minimum of 6 credits and Maximum of 10 credits
- Business electives - Minimum of 6 credits and Maximum of 10 credits
- Bridge/System electives - Minimum of 8 credits and maximum of 12 credits
    - Bridge Course – 4 credits (only if candidate doesn't pass exemption exam)
- Open electives – Minimum of 0 credits and Maximum of 8 credits
- Product Development Project – 16 credits
- Industry/Institute Seminar – 0 Credits

Project Work:

- The product development project will be an integrated effort to create a market connected product or solution. May lead to a startup or a product that can create some social impact. For tech professionals from industry, the project may lead to a very relevant product that can be used in government or social sectors. And for startup aspirants, the thesis may lead towards the first market MVP of their startup idea.

- Product Development project completed in Third & Fourth semesters with 16 credits each will follow Satisfactory(S)/Unsatisfactory(X) grading scheme without grade points.

Academic Performance:

- A student should complete the requirements with a minimum CGPA of 6.00 to receive the M.Tech degree.

Residency Requirements:

- **Full-time students:** Students will have minimum of 4 semesters and maximum of 6 semesters to complete the graduate requirements, failing which they will be terminated from the program.

- **Industry Students (Part time mode):** M.Tech program may be pursued concurrently with your work. Currently, this option is available for Industry / Self sponsored candidates & Startups.

Fees:

- The student must pay full-time post-graduate fees for the first 4 semesters of study. The fees will be pro-rated to the number of credits registered for thereafter according to the institute's policies.

- Industry students (selected via the self-sponsored and startup mode) have the option of pursuing the program in a part-time student. They can choose to enroll for lesser credits compared to full-time student. The students may choose a minimum of 8 credits up to a maximum of course load equivalent to a full-time student depending on their current professional commitment.

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M.Tech in VLSI & Semiconductor Technologies
Academic Regulations (Highlights)

(Applicable from 2026 batch)

Credit Requirements:

- Minimum credits required for graduation is 66. Each semester, every student must register for at least 12 credits and at most 20 credits, which can include full or half courses
- Every student must complete 36 area elective credits which may also include 8 Independent Study credits.
- ECE area elective courses belong to 3 categories- (1) VLSI Design (2) Embedded Systems & AI, and (3) Semiconductor Technologies. A student must complete 1 course from each of these 3 categories, and at least 3 courses in one of these three categories.
- The list of courses in these categories is given below. These courses are usually offered at least once every year.
- Every student must register for 4 credits of open electives, which include any 400+ level elective offered in the institute. Open electives include bouquet core electives.
- Every student must register for 2 credits of summer project, which include works assigned by M.Tech Project Advisor or Industrial internship under the mentorship of the Advisor.
- Every student must register for 24 credits of M.Tech Project, which include 8 credits in the third semester and 16 credits in the fourth semester.

Area Electives:

Area Electives are broadly classified under following three categories covering various domains such as Digital, Analog, RF, Mixed-Signal, EDA, Computer-Architecture, Nanotechnology, Device physics, Device fabrication and Flexible Electronics.

A student must complete 1 course from each of these 3 categories, and at least 3 courses in one of these three categories.

Category-I: VLSI Design: Analog IC Design, Digital VLSI Design, Principles of Semiconductor Devices, CMOS References and Regulators, CMOS Oscillators

Category-II: Embedded Systems & AI: Digital VLSI Design, Advanced Comp Architectures, Compilers, Hardware for AI, Software for performance

Category-III: Semiconductor Technologies: Principles of Semiconductor Devices, Advanced Semiconductor Devices, Flexible electronics, Instrumentation & Devices, Nano Fabrication Lab, Microelectronics Simulation Lab, III-V Compound Semiconductors, Nano Technology, MEMS and THz Technology, Optoelectronics

M.Tech Project Work:

Every student must register for 24 credits of M.Tech Project, which include 8 credits in the third semester and 16 credits in the fourth semester.

Academic Performance:

- A student should complete the requirements with a minimum CGPA of 6.00 to receive the M.Tech degree

Residency Requirements:

- Full-time students will have minimum of 4 semesters and maximum of 6 semesters to complete the graduate requirements, failing which they will be terminated from the programme

Fees:

- The student has to pay full-time post-graduate fees for the first 4 semesters of study. The fees will be pro-rated to the number of credits registered for thereafter according to the institute's policies.

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